

HabitBuddy

Group 10: GP4 Final Report

CS 5340 – Computer/Human Interaction

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1. Problem

Many people struggle to build and maintain habits consistently. HabitBuddy is a habit-tracking system designed to make habit formation emotionally engaging and sustainable through fast logging, clear progress visibility, and a growth companion pet whose mood and progression reflect a user's habit consistency. The intention-behavior gap is the primary reason people fail to sustain new habits. HabitBuddy addresses this through a companion-based reward model.

Target Users

In GP2 we worked on user personas to capture the user age range (18-30 year) : a goal-oriented student cramming for technical interviews, a busy engineer trying to build a reading habit around an unpredictable schedule, a high schooler who needs visible progress to stay focused, and a fitness-minded professional tracking structured workouts toward a race. Across these users, four core tasks emerged: defining and configuring a new habit with realistic frequency and goals, logging daily completions quickly enough that the interaction never feels like a chore, tracking progress over time through clear visual feedback and customizing a companion pet using earned rewards to strengthen emotional investment.

2. Design

Final Design Overview

The final interface centers these tasks around three main areas: Home, Habits, and Progress. The Home screen emphasizes the companion and the immediate reward structure. Users can see today's habits, mark them complete, and view the pet's current state, XP, and available points. The Habits screen supports management tasks such as adding, editing, deleting, and reviewing the current habit list. The Progress screen summarizes current streak, total points, level, and daily consistency. The interface uses warm, soft colors and rounded card-based layouts to make the system feel supportive rather than clinical. We deliberately chose a playful visual language because

it reinforces the motivational logic of the product: HabitBuddy should feel like something users want to return to, not something that only reminds them of unfinished work.

Key Design Decisions and Alternatives

Design Decision motivated by findings from the paper prototype and the heuristic evaluation.

Decision 1: companion-centered home screen During working on paper prototype we chose a data-first dashboard. An alternative we considered was a more conventional productivity layout with charts, percentages, and a dense task list as the first screen. That alternative would have made analytics more visible, but it weakened the main differentiator of the concept. The final version keeps progress metrics available while making the bun the emotional anchor of the experience.

Decision 2: edit/complete separation (enlarged pencil) This came from both evaluations. This issue first appeared in paper prototype testing, where participants clicked the habit card itself rather than the small edit control on the side, we changed it to show the edit message at top and reappeared during heuristic evaluation when evaluators flagged weak edit discoverability.

Decision 3: calendar picker instead of text field This came from heuristic evaluation. This change was motivated directly by the heuristic evaluation evaluator who noted that a calendar should appear rather than the keyboard and no participant in the paper prototype sessions had encountered the date field in the earlier design.

Decision 4: removing the Customise tab / making pet tappable This change came directly from paper prototype testing, where every participant instinctively tapped Bun expecting customization rather than navigating to the separate tab. No heuristic evaluator flagged this because the change had already been made before the heuristic evaluation session.

Paper Prototyping: What We Learned

We recruited a total of four participants for testing of the paper prototype. All participants were selected to represent the target user group, and we conducted the prototype testing in a controlled environment. We got creative in our paper prototyping process and focused on integrating the pet into our UI design. The clearest learning from the paper prototype process was that users do not read the interface they act on it instinctively and any mismatch between what looks tappable and what actually does something will cause immediate confusion.

After the observations, we connected the pet on the Home screen to the Customize sheet and removed the separate Customise tab, matching the mental model users already had. We turned the frequency day preview into the actual selector so the interactive and visual elements were the same object. We consolidated the progress view around a single consistent model (filling the bar) rather than mixing daily and cumulative displays. We reduced the bottom navigation from four tabs to three, cutting the number of screens users had to cross to complete common actions. We also made the edit affordance more noticeable.

What worked well. The pet-based reward concept was understood immediately and received positive emotional reactions. Participants found the combination of habit tracking and pet growth interactive and helpful. Several participants commented on Bun unprompted during their sessions. Post-test interviews confirmed that participants wanted a smoother version of this concept rather

than a fundamentally different one. This gave us confidence to invest in the companion interaction rather than reconsider the core premise.

Heuristic Evaluation: What We Learned

Five evaluators completed all four tasks during the in-class heuristic evaluation and rated the interface against Nielsen's ten heuristics on a 1–4 scale (1 = catastrophic, 4 = cosmetic). All five evaluators completed Tasks 2, 3, and 4. Task 1 (Create a New Habit) was completed by four evaluators fully and by one evaluator partially, who noted that creation confirmation was missing. No evaluator reported a catastrophic failure that prevented task completion.

Issue 1: Weak edit discoverability. *Severity: major* The text “tap to edit habit wasn’t useful”. This change was prioritized because it affected visibility and affordance principles. We corrected it in final design.

Issue 2: Weak confirmation after habit creation. *Severity: minor to major*, depending on task context. The system returned users to the list after creation, but the feedback was easy to miss. Evaluators were not always sure whether the habit had been saved or whether the transition was simply taking them back. We addressed this by adding a dedicated “Habit Created!” success state. The new state provides explicit system feedback before returning users to the list, improving visibility of system status and reducing uncertainty.

Issue 3: Date fields implied typing instead of selection. *Severity: major.* Users expect date entry on mobile to use a picker, calendar, or otherwise constrained interaction. We resolved this by replacing the text-like behavior with calendar popup.

Read the Appendix at the end to see the screenshots of the changes we made to the design.

3. Evaluation

Proposed Evaluation Methods

For the final prototype interface of HabitBuddy, given more time, we would pursue the following evaluation approaches:

- **Verify that usability fixes work:** Run another round of user testing with the final prototype, using the same tasks. Measure whether the UI is intuitive after the fixes.
- **Assess real-world engagement:** Give 5–8 participants access to the app for one week and observe whether they return daily. A habit tracker that people use once and abandon has failed its core purpose, regardless of how clean the UI is.
- **A/B test the progress screen:** We discussed several alternative representations for daily streaks and progress visualization within the group but did not pursue them due to implementation complexity. A/B testing would help determine which representation creates stronger recall.

Proposed Next Steps

- Deploy the prototype as a hosted web app or create an app to enable more rigorous retention testing of whether the pet companion sustains habit behavior over time.

- Do a structured task-based usability study with five to eight participants from the target demographic. We would ask users to create a new weekday habit, mark a habit complete from the Home screen, edit an existing habit, and customize the pet after earning points. We would measure task success, hesitations, incorrect taps, and time on task, then pair those measures with short post-task interviews about clarity, motivation, and reward.

Remaining Usability Issues

- **Progress bar lacks granularity:** The combined 28-day progress bar does not convey enough detail to the user. Adding filters or an interactive pie chart would let users explore their progress more meaningfully.
- **Reminder notifications:** We thought of adding a fun element by creating fun reminders based on the tone the user chose, we didn't get to flesh this idea out and implement it.
- **Pet growth feedback:** Bun visually scales up as it levels, but there is no celebration moment. A level-up celebration screen when pet reaches a new level would reinforce the reward loop.

4. Reflection

The project reinforced that good interaction design comes from repeated cycles of making, testing, and simplifying. The strongest parts of the final prototype were not the ones we got right immediately; they were the ones that improved after paper prototyping and heuristic evaluation exposed mismatches between our assumptions and user expectations.

What We Would Do Differently

On prototype fidelity decisions: If we repeated the process, we would decide earlier which interactions needed to feel dynamic and which could remain static. Some of our later work involved retrofitting interactions that evaluators had already encountered in more static form. The GP3 paper prototype already showed us that users wanted to manipulate the bunny, habit cards, and scheduling controls directly, and we could have responded to that sooner.

On evaluation timing: We would bring fresh evaluative feedback into the form flows earlier. Several of the strongest issues, especially around discoverability and date input, were not conceptually difficult problems. They simply needed to be seen by outside users earlier in the semester.

On form scope management: We would manage scope more aggressively from the start. Our instinct was to include many potentially useful settings so the concept would feel complete, but both the paper prototype and the heuristic evaluation showed that completeness and clarity are not the same thing.

On the companion metaphor: We would continue exploring the companion-based reward model, but with more deliberate testing of how playful the interface should be relative to its productivity functions. That balance is central to HabitBuddy's identity, and it deserves explicit evaluation rather than relying only on intuition.

Appendix: Post-Heuristic Evaluation Design Changes

The following screenshots document the four design changes made to the HabitBuddy prototype in response to findings from the heuristic evaluation conducted on April 3rd, 2026.

Change 1 — Edit Affordance Discoverability

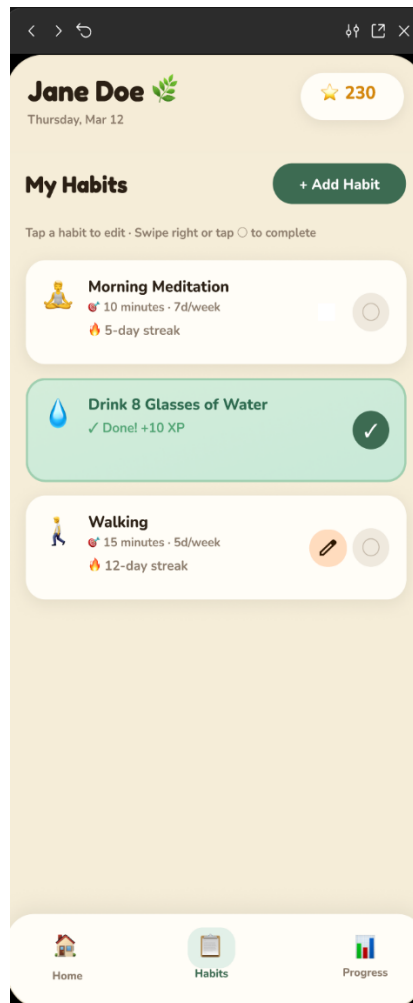


Figure 1. My Habits screen (post-iteration) — pencil icon is now visible alongside the completion circle, making the edit action independently discoverable.

Change 2 — Confirmation After Habit Creation

The original design returned users to the My Habits list after saving a new habit, with a toast banner ('New habit added!') that appears immediately on the list which was a minor visibility problem and now we have a full-screen Habit Created success state that displays for approximately 1.5 seconds before returning to the list, featuring Bun to reinforce the companion reward loop.

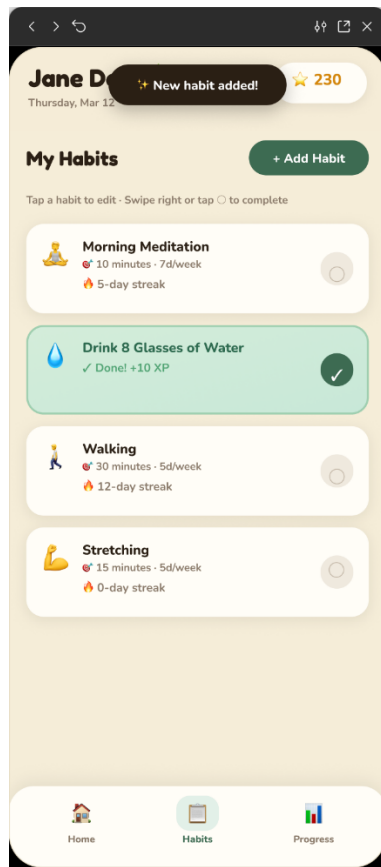


Figure 2a. Pre heuristic evaluation Toast banner 'New habit added!' appears at the top of the list immediately after saving.

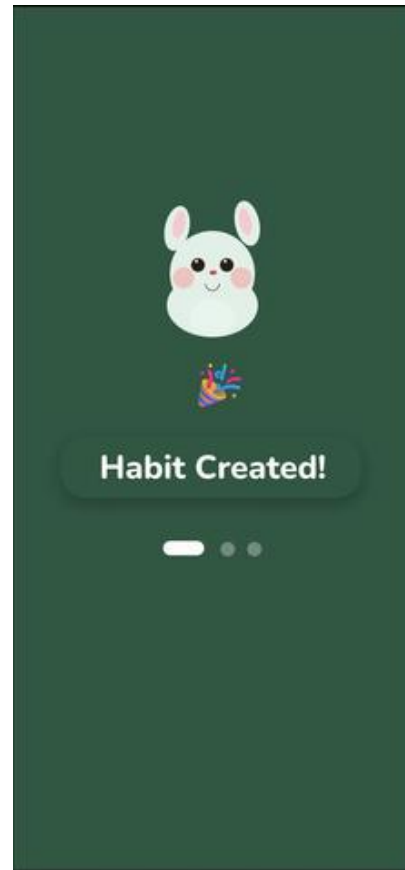


Figure 2b. Post Heuristics evaluation: Full-screen 'Habit Created!' success state with Bun, shown before returning to the list.

Change 3 — Calendar Picker Replaces Text Field

In the pre-evaluation design, the date input field looked like a standard text field, which caused the mobile keyboard to appear when tapped. The updated design replaces the text field behaviour with a calendar overlay that appears when the date field is tapped, matching the standard mobile date selection pattern.

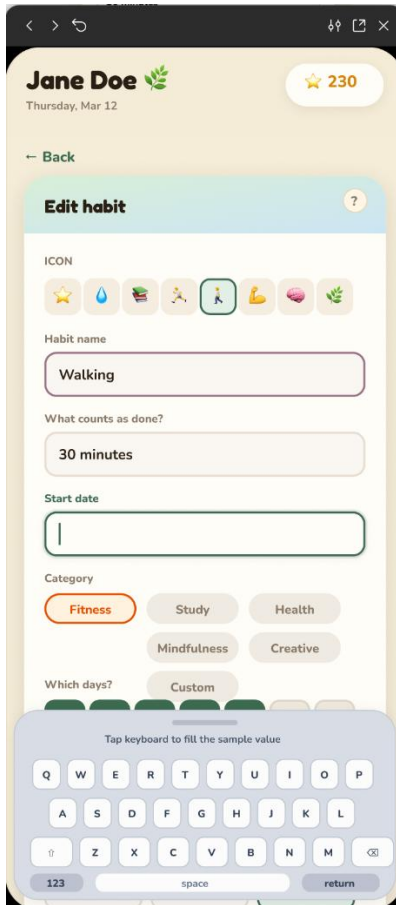


Figure 3a. Before tapping the date field triggers the keyboard.

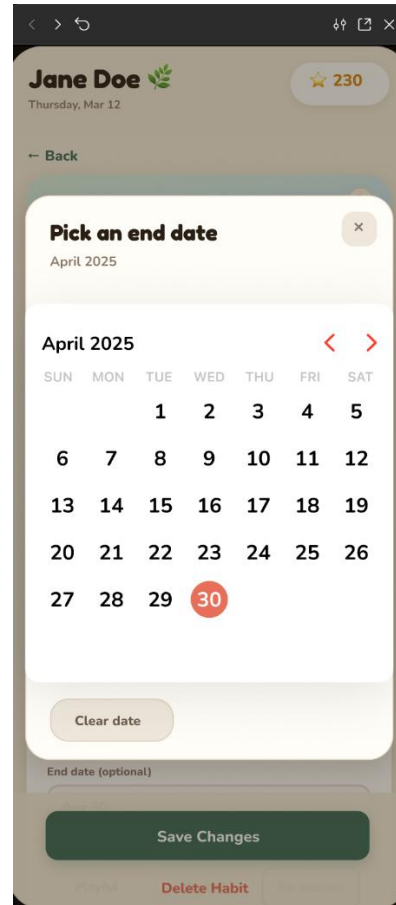


Figure 3b. After tapping the date field opens a calendar overlay picker.